

3.2.9 Design for manufacture and project management

Planning for accuracy and efficiency

Content	Potential links to maths and science
Students should be aware of, and able to discuss and demonstrate, the importance of planning for accuracy when making prototypes and making recommendations for small, medium and large scale production.	

Quality assurance

Content	Potential links to maths and science
Students should be aware of, and able to discuss and demonstrate, the procedures and policies put in place to reduce waste and ensure manufactured products are produced accurately and within acceptable tolerances, including quality assurance systems including Total Quality Management (TQM), scrum, Six Sigma and their applications to specific industrial examples including critical path analysis.	

Quality control

Content	Potential links to maths and science
Students should be aware of, and able to discuss and demonstrate, quality control, including: • the monitoring, checking and testing of materials, components, equipment and products throughout production to ensure they conform to acceptable tolerances • specific quality control methods including the use of 'go-no go' gauges, laser or probe scanning and measuring • use of digital measuring devices such as vernier callipers and micrometers • non-destructive testing such as x-rays and ultrasound.	